QUICK START & SAFETY GUIDE · SKU CQA240229G16

From a Bare Pi to 16 Live Channels

Seven steps from a Raspberry Pi to live 16-channel EEG in the browser. PiEEG-16 is PiEEG Server's default device, so there is nothing to configure after install.



WARNING Battery only.

PiEEG-16 has **no isolation circuitry**. The Raspberry Pi, the shield and everything cabled to them (monitor, keyboard, mouse) must run from a **5 V power bank** while electrodes are on a person. PiEEG-16 is **not a medical device**.

01 SAFETY RULES — READ BEFORE PUTTING ELECTRODES ON ANYONE

#	Rule
1	**Battery only.** Pi, shield and everything cabled to them run from a 5 V power bank.
2	**No mains connection of any kind:** no wall charger, no desk supply, no USB or HDMI cable to a mains-powered computer, monitor or TV.
3	**No diagnosis.** Do not use PiEEG-16 to diagnose, monitor or treat any medical condition.
4	**Implanted devices.** Not on a person with a pacemaker or other implant without a physician's approval.
5	**Skin.** Clean, intact skin only. Stop if it becomes irritated; clean off gel after each session.
6	**Exposed electronics.** No case: handle by the edges, use anti-static precautions, keep liquids away.

Power while recording

Setup	While electrodes are on	Why
Pi + shield on a 5 V power bank	SAFE	No path to the mains.
Monitor powered by the power bank or the Pi	SAFE	Still fully on battery.
Laptop on battery, dashboard over Wi-Fi	SAFE	No cable to the Pi or the person.
Pi on a wall charger	NEVER	Connects the person to the mains through the electrodes.
Monitor plugged into a wall socket	NEVER	The HDMI cable links the mains to the Pi.
USB / Ethernet to a mains-powered computer	NEVER	Same reason.

02 WHAT YOU NEED

Item	Requirement
PiEEG-16	Included
Raspberry Pi	Pi 4 or Pi 5, microSD 16 GB+ with Raspberry Pi OS (64-bit, desktop)
5 V power bank	5 V, 3–3.5 A, up to 10,000 mAh; short, thick USB cable
Electrodes	16 dry or wet Ag/AgCl with 2.54 mm connectors + 2 ear clips
EEG cap	16-electrode 10-20 cap
Monitor, keyboard, mouse	Monitor powered from the bank or the Pi; wired keyboard and mouse
Gel	Only for wet electrodes

03 SETUP IN SEVEN STEPS

1 Gather the parts

PiEEG-16, Raspberry Pi 4 or 5, charged power bank, 16 electrodes and 2 ear clips, cap.

2 Prepare the Raspberry Pi

Flash Raspberry Pi OS (64-bit, with desktop) with Raspberry Pi Imager, boot once and connect to the internet for installing software.

3 Mount the shield

Shut down the Pi and unplug the power bank. Line up pin 1 with pin 1 on the 40-pin header and press down evenly until every pin is seated.

4 Connect the electrodes

Channels 1–8, **REF** and **BIAS** on the left header; channels 9–16 on the right header. REF to one earlobe, BIAS to the other. See PE16-PIN-004.

5 Power on from the battery

Connect the battery-powered monitor and wired keyboard/mouse, then plug the power bank into the Pi. The LEDs on the shield light up.

6 Install and start PiEEG Server

TERMINAL ON THE PI

```
curl -sSL https://raw.githubusercontent.com/pieeg-club/PiEEG-server/main/install.sh | bash
sudo reboot              # first time only: enables SPI
pieeg-server doctor      # check SPI, GPIO and settings
```

PiEEG-16 is the default device: the service starts streaming 16 channels on every boot.

7 Check the signal

Open **http://localhost:1617** on the Pi (or http://raspberrypi.local:1617 from a battery-powered laptop) and click **Connect**. You should see 16 channels.

04 FIRST SIGNAL CHECKS

Test	What you should see
Sit still for 30 s	All 16 channels settle. Flat lines or huge values mean a loose electrode.
Close the eyes for 10 s	Alpha (8–13 Hz) rises on O1, O2, P3, P4, then drops when the eyes open.
Blink a few times	Large slow spikes on Fp1 and Fp2.
Clench the jaw	High-frequency bursts, strongest on T7 and T8 — muscle, not brain.

05 TROUBLESHOOTING

Power and hardware

Problem	Fix
LEDs on the shield are off	Check the power bank is on and charged, and the shield is seated on all 40 pins.
Pi restarts / lightning bolt	Power bank too weak. Use 5 V / 3 A and a short, thick cable.
Shield off by one pin	Shut down, remove and re-seat; pin 1 must match pin 1.

SPI, GPIO and data

Problem	Fix
`/dev/spidev0.0` or `0.1` not found	SPI is off: sudo raspi-config → Interface Options → SPI → Enable, reboot.
Permission denied on SPI/GPIO	`sudo usermod -aG spi,gpio $USER`, then log in again.
GPIO chip not found	Run `gpiodetect`; on a Pi 5 use pinctrl-rp1. Edit the script or pass --gpio-chip.
Device busy on GPIO 19, 26 or 13	PiEEG Server holds the pins: sudo systemctl stop pieeg-server.
Channels 1–8 work, 9–16 zero or random	Chip 2 is not read: check DRDY 2 (GPIO 13), CS on GPIO 19 and /dev/spidev0.1.
Occasional single-sample spikes	SPI glitches: lower the SPI clock (e.g. 1 MHz); clean files with Remove_Spikes.py.
Dashboard shows 8 channels	Start with --device pieeg16, or without --device.

Signal

Problem	Fix
All 16 channels noisy at once	Check the **REF and BIAS ear clips** first.
One channel flat or very large	That electrode is not touching the skin.
Strong 50/60 Hz hum	Something is on the mains — unplug every charger, run the monitor from the battery.
Signal jumps when someone moves	Cable movement — bundle the 16 leads and keep them still.
Alpha does not rise with eyes closed	Check O1, O2, P3, P4 touch the scalp; eyes closed 10 s in a quiet room.

06 RELATED DOCUMENTS

PE16-DS-001 Product Datasheet	**PE16-MD-002** Mechanical & Dimensional Drawing	**PE16-PCB-003** PCB & Manufacturing Specification
PE16-PIN-004 Connector Pinout & Electrode Guide	**PE16-SW-006** Data Interface & Software Guide	**PE16-ENC-007** 3D-Printed Enclosure Guide